

Ali Meski

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EDUCATION

Bachelor of Engineering, Systems and Computing Engineering (Co-op)

2024–Present

University of Guelph, Guelph, Ontario

- Dean's Honours List – Winter 2025; Entrance Scholarship of \$2,000 for 93% admission average
- Relevant Courses: Data Structures, Digital System Design, Object-Oriented Programming

SKILLS

Data Science & ML: Exploratory data analysis, predictive modelling, model evaluation, statistical analysis, feature engineering

Python Ecosystem: PyTorch, scikit-learn, NumPy, pandas, Matplotlib, spaCy, tiktoken, FastAPI, Docker

Data & Pipelines: SQL, ChromaDB, data pipeline design, text extraction, data cleaning and validation, result reporting

Other Tools: Git, MATLAB, VS Code, PyCharm, Google Cloud Platform, Microsoft Excel

EXPERIENCE

Robotics Research Assistant

May 2025 – Present

University of Guelph, Dept. of Electrical and Computer Engineering

Guelph, Ontario

- Building a **predictive assistive robot** in Python to support individuals with cognitive disabilities, using motion sensor data to detect behavioural deviations and trigger real-time assistive cues.
- Designed a **modular software architecture** with clean separation between data ingestion, anomaly detection, and alert output layers, deployed on a Raspberry Pi connected to an inertial motion sensor.
- Preprocessing motion sensor datasets and developing explainable anomaly detection models using statistical thresholds and clustering to identify when a user may need assistance.
- Integrating local LLMs (Llama 3.2 3B) and object detection pipelines (YOLO) to enable context-aware decision making, evaluating performance across cloud and local deployment environments.

PROJECTS

LLM-Powered Medical Research Paper Summarizer | *Python, FastAPI, spaCy, Docker*

- Built an end-to-end **medical document processing pipeline** using PyMuPDF, pdfplumber, spaCy, and tiktoken to extract and clean scientific text, mirroring the data ingestion and feature engineering work involved in processing clinical records.
- Designed a **map-reduce LLM pipeline** with token-aware chunking, fallback logic, and retries across Claude and GPT APIs, demonstrating experience building robust, production-grade data and modelling pipelines.
- Implemented **automated model evaluation** using ROUGE and BERTScore metrics with structured performance logging to rigorously assess output quality and track accuracy across iterations.

Endangered Species Visualizer | *React, TypeScript, Tailwind CSS, Mapbox API*

- Collected, cleaned, and consolidated 250+ records from dispersed datasets, performing exploratory analysis to identify patterns, quality issues, and key features for filtering and geospatial visualization.
- Built interactive dashboards and reporting tools to communicate complex analytical findings clearly to public and academic audiences, demonstrating experience translating data outputs into usable, interpretable results.

Gameboy Emulator | *C, SDL2*

- Engineered a fully working Game Boy emulator in **C**, replicating CPU, memory, graphics (PPU), audio (APU), and input subsystems with cycle-accurate instruction decoding for full ROM compatibility.
- Architected a modular system design with **SDL2** for real-time graphics rendering, enhancing maintainability and enabling scalability for additional emulator features.

Sock Sensei | *Kotlin, Android SDK*

- Developed an Android application in **Kotlin** with multi-activity navigation, custom UI components, and a responsive user experience built around Android architecture best practices.
- Implemented local data persistence with structured storage for reliable state management, and applied dependency injection and unit testing to ensure code quality across iterative development.